ClipSAT : Math for AI & Data Science

Unit 3.01: Gradient Descent & Optimization

Name: _____

Q1. In gradient descent, a weight currently has value w = 20 . If the gradient of the loss is frac{partial L}{partial w} = 8 and the learning rate is eta = 0.2 , find the updated weight w' .

[ ] A: 12.00
[ ] B: 21.60
[ ] C: 18.40
[ ] D: 19.80

Q2. In gradient descent, a weight currently has value w = 30 . If the gradient of the loss is frac{partial L}{partial w} = 12 and the learning rate is eta = 0.3 , find the updated weight w' .

[ ] A: 18.00
[ ] B: 33.60
[ ] C: 26.40
[ ] D: 29.70

Q3. In gradient descent, a weight currently has value w = 60 . If the gradient of the loss is frac{partial L}{partial w} = 24 and the learning rate is eta = 0.6 , find the updated weight w' .

[ ] A: 36.00
[ ] B: 74.40
[ ] C: 45.60
[ ] D: 59.40

Q4. Consider the scalar loss function Lw = 2w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 2 .

[ ] A: nabla L2 = 16
[ ] B: nabla L2 = 8
[ ] C: nabla L2 = 0
[ ] D: nabla L2 = -4

Q5. Consider the scalar loss function Lw = 3w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 4 .

[ ] A: nabla L4 = 16
[ ] B: nabla L4 = 24
[ ] C: nabla L4 = 32
[ ] D: nabla L4 = 4

Q6. Consider the scalar loss function Lw = 4w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 6 .

[ ] A: nabla L6 = 40
[ ] B: nabla L6 = 48
[ ] C: nabla L6 = 56
[ ] D: nabla L6 = 16

Q7. Consider the scalar loss function Lw = 5w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 8 .

[ ] A: nabla L8 = 72
[ ] B: nabla L8 = 80
[ ] C: nabla L8 = 88
[ ] D: nabla L8 = 32

Q8. Consider the scalar loss function Lw = 6w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 10 .

[ ] A: nabla L10 = 112

[ ] B: nabla L10 = 120

[ ] C: nabla L10 = 128

[ ] D: nabla L10 = 52